

SQL LAB 0

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Aim: To study SQL commands, their uses, syntax, and install MySQL for database management experiments.

SQL stands for structured query language. It lets u access and manipulate databases

The databases are in relation format, or we can say the data stored in the relational tables

SQL can execute queries, retrieved data, insert, update, delete and even create new records

Sql can be used in your websites or rather we can say it can be connected to your software you made for example using python **MySQL** connector, you can connect your sql database server to your python backend,

SQL is based on **rdbms**(relational data base management system) and it is also basis for all modern database systems most common query or the simplest is to retrieve data from table using sql query eg: select * from student; where * returns with all the data present in the table

sql keywords or the query keywords are not case sensitive that is
select is as same as SELECT semicolon(:) is supposed to be added
at end of every sql queries **Some important sql command**

- **select:** extracts data from aa database or a table
 - **update:** updates data into database
 - **delete:** deletes data from database
 - **insert into:** inserts new data/values into the database
 - **create database/ create schema:** creates new database
 - **alter database:** modifies database
 - **create table:** creates new table
 - **alter table:** modifies the table
 - **drop table:** deletes the tables create
 - **index:** creates an index(search key)
 - **drop index:** deletes the index
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SELECT STATEMENT

syntax: select <column/columns> from <table>;

eg: select student_id from student_data_table;

select * from student_id;

SELECT DISTINCT STATEMENT

select distinct statement is used to return only unique values it only display unique values from a column which may have duplicate values

synatax: select distinct column/columns from table;

eg: select distinct country from customers;

SELECT COUNT(DISTINCT)

eg: select count(distinct country) from customers;

eg: select count(*) as distinctcountries from (select distinct from customers);

SELECT WHERE CLAUSE

the where clause is to filter records or used to extract only those records which fulfill a specific condition

syntax: select column/columns from table_name where condition;

eg:select * from customers where customerID =1;

where has diff operations

=,>,<,<=,>=, <>, between, like, in, starts_with, ends_with etc.

UPDATE STATEMENT

syntax: update table_name set column1=value1, , column2=value2, where condition;

Eg: update customers set contactname='john doe', city='frankfurt' where customerID=1; update customers set contactname='juan' where country='mexico'
****skipping where clause in UPDATE clause results into updating of all records****

SQL ORDER BY

the order by keyword is used to sort the result-set in ascending or descending order and is set to ascending by default

syntax: select column/columns from table_name order by column;

eg: select * from products order by price;

select * from products order by price desc;

select * from customers order by country asc, customername desc;

SQL DELETE STATEMENT

synatax:

delete from table_name where condition; eg:

delete from customers where customername='john doe';

to delete a whole table:

syntax: drop table table_name;

eg: drop table customers;

SQL INSERT INTO STATEMENT

syntax: insert into table_name(column1,column2,column3) values(value1,value2,value3);

if you want to insert values for all the columns of the table, u can omit the column names and enter the values in the order of the columns syntax:

insert into table_name values(value1,value2,value3);

eg:

insert into customers values ('cardinal','tom b erichsen', 'Skagen 21', 'stavanger', '4006', 'norway');

CREATE DATABASE STATEMENT

eg: create database

testDB;

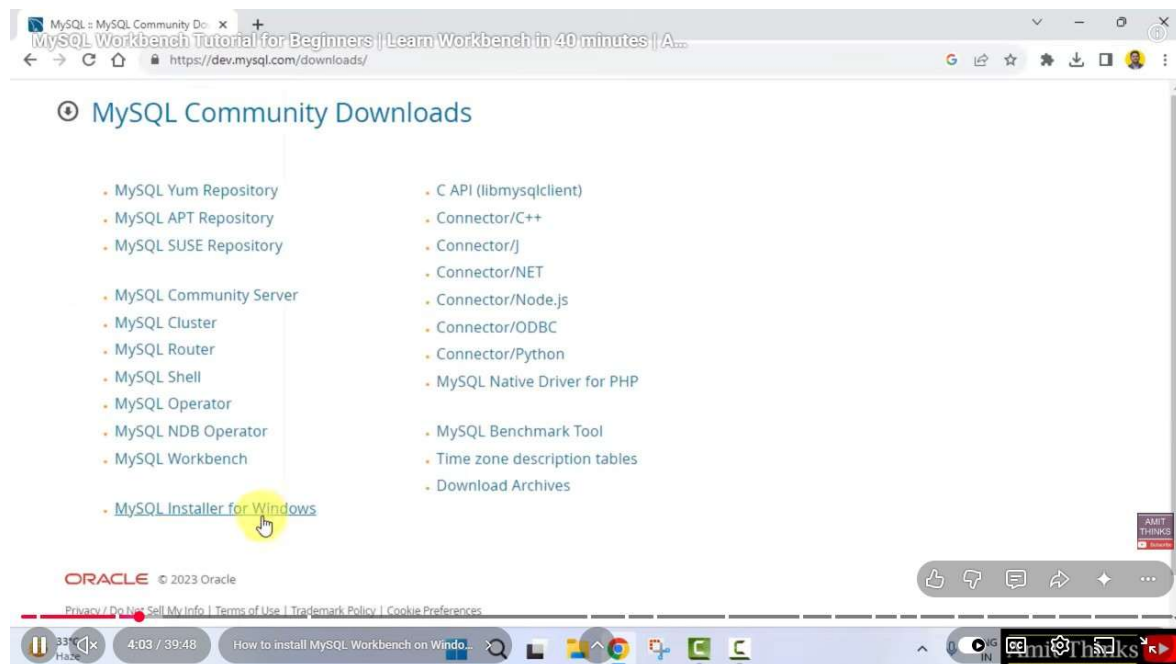
CREATE TABLE STATEMENT

eg:

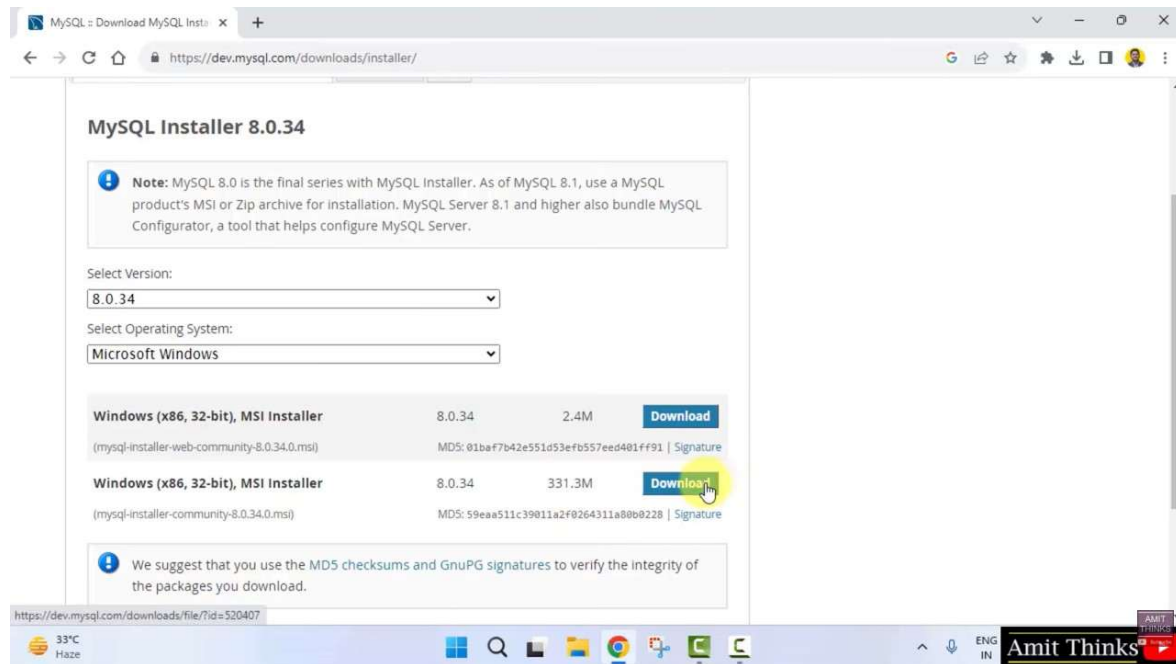
create table tableTEST;

Installation Procedure for MySQL

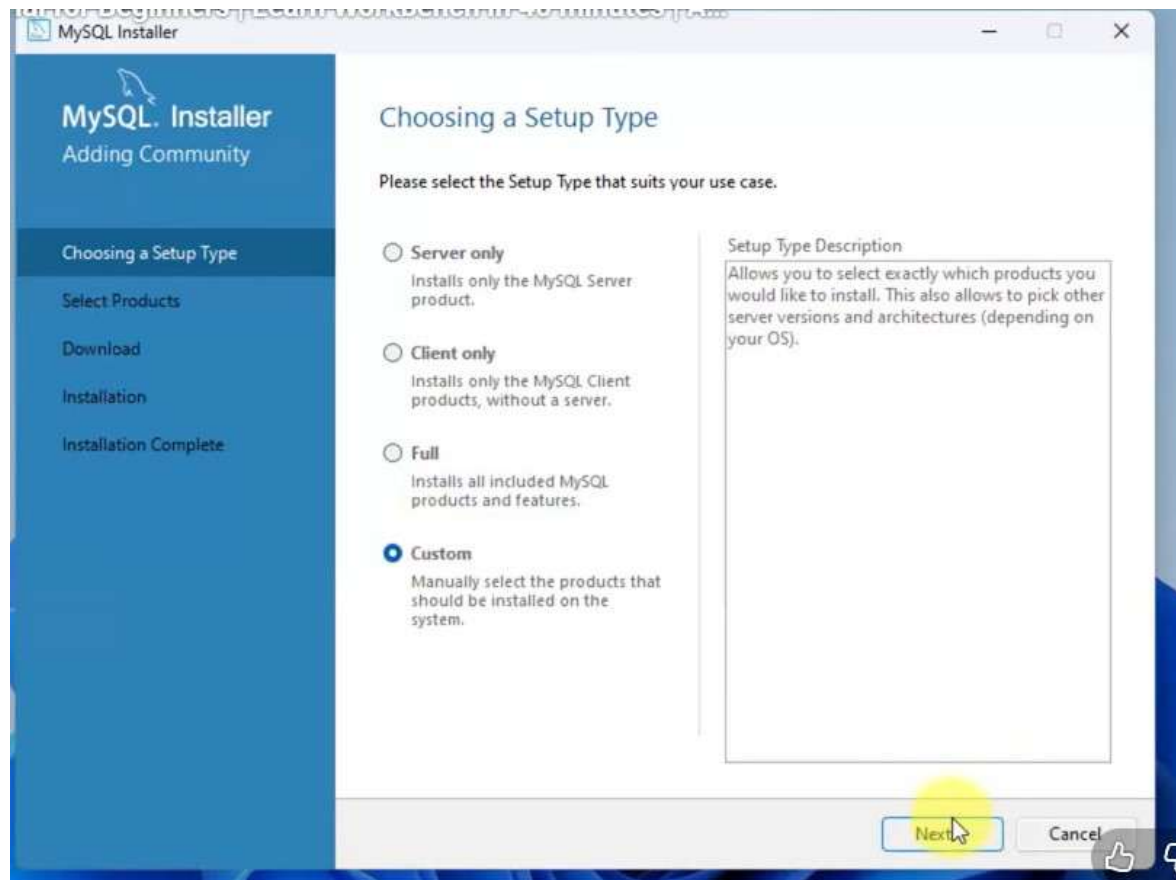
1. Go to <https://www.mysql.com/> and navigate to download page and click on MySQL for Windows



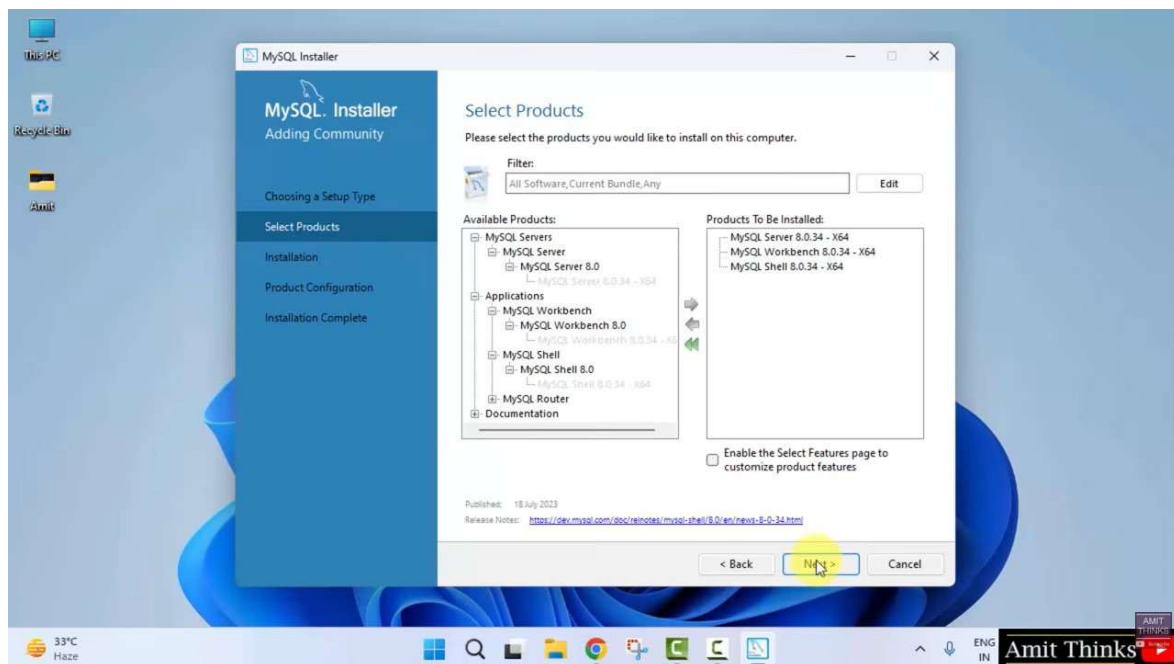
2. Select the preferred Installer and click on Download



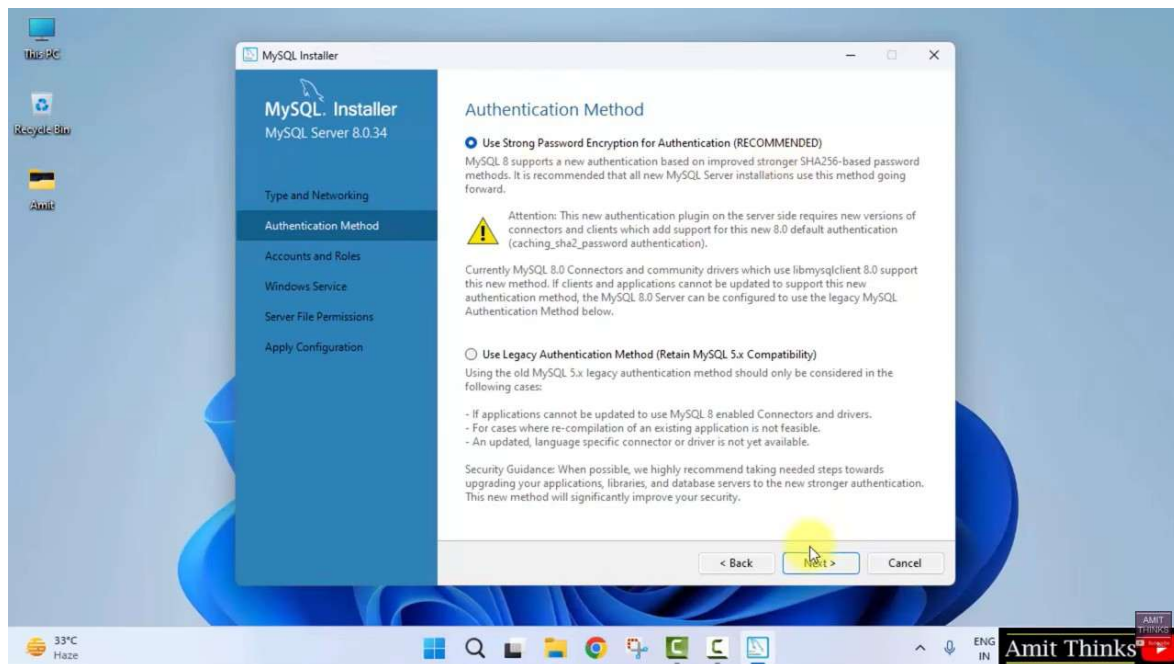
3. After Running the Installer, Select Custom Setup and click on Next



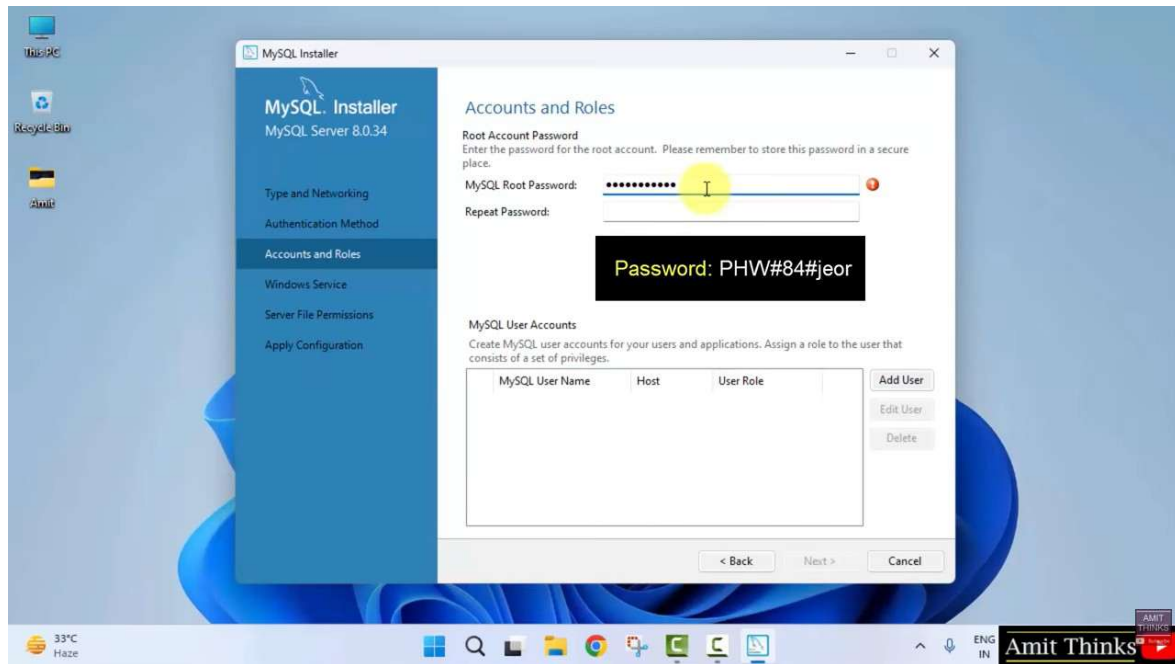
4. Select the preferred Products to be Installed. Preferably Server, Workbench and MySQL Shell



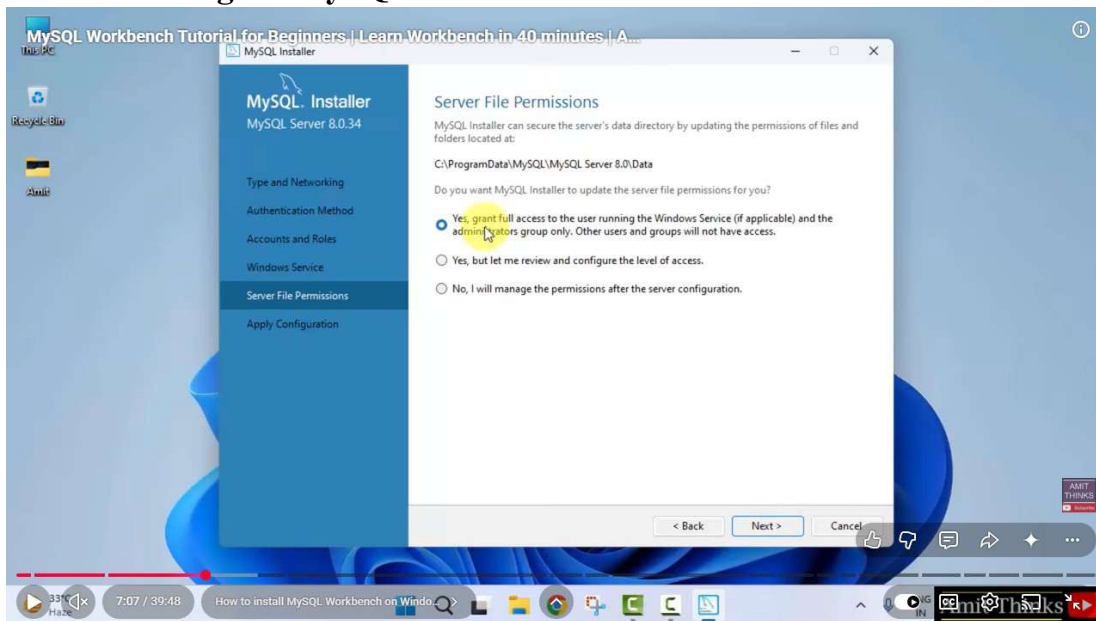
5. Select the Port You want for the Server and click on next



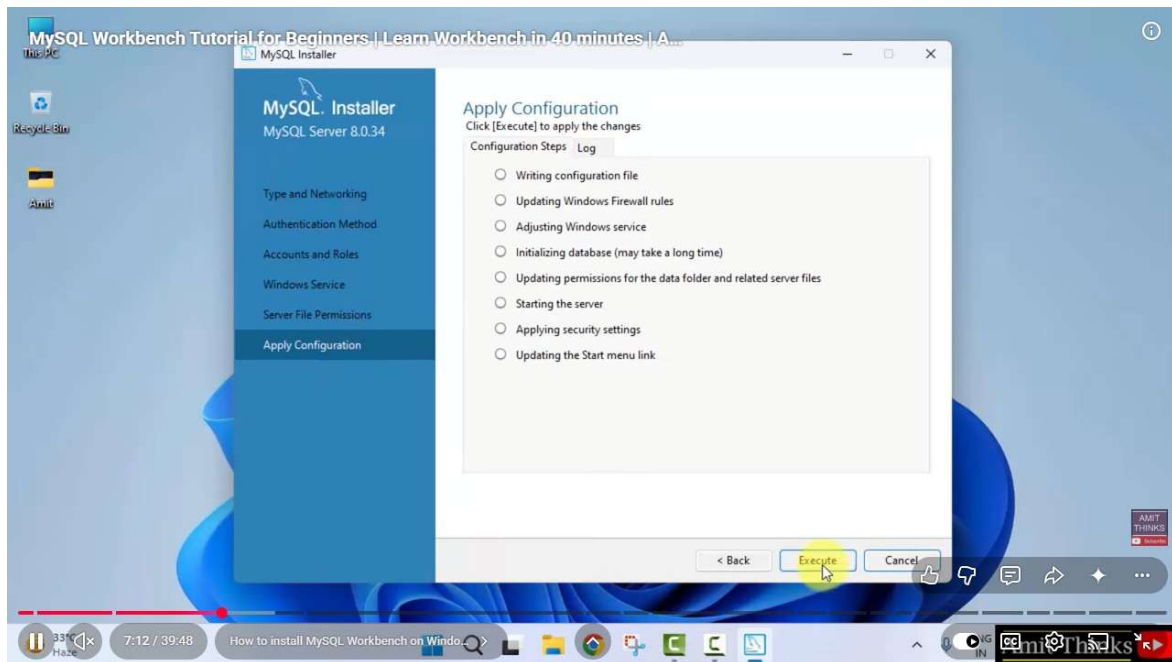
6. Set Your Password for the Root account



7. Select Configure MySQL server as a Windows Service and Click Next



8. Click on Execute to complete the action



9. Click on Finish to Complete Installation

